A Local Hybrid Biomedical Image-Analysis Pipeline

VLM Description, Classical Segmentation and U-Net for Fluorescence Nuclei Imaging

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GitHub Link: https://github.com/samiaarif199510/Assignment3_AI_imaging

Overview. This report introduces a small and traceable biomedical image-analysis workflow for fluorescence microscopy nuclei images. It consists of four key components: exploratory data analysis (EDA) and an LLM-generated description for an exploratory multimodal vision-language model (VLM); classical Otsu thresholding followed by a numbers-first LLM description; LLM interpretation of a structured description of the segmentation masks generated by a U-Net, evaluated based on Dice and IoU metrics on the unseen test set; and a hybrid pipeline that transforms the U-Net segmentation masks into structured JSON records and then into narrative descriptions by LLM on the unseen test set. Extension experiments encompass a loss-function ablation and corruption-robustness analysis. Finally, the report offers direct answers to the five questions listed above. All LLM-based components in this study were run on a live local Ollama server, with MiniCPM-V for vision-language analysis and Llama 3.1 for text-based analysis.

1. Data Preparation and Multimodal LLM Description (Task 1)

The data set (fluorescence microscopy images of nuclei) had been loaded and converted to grayscale and resized to a 256×256 image. The blue fluorescence channel contains almost all of the foreground, which allows the use of the grayscale conversion to keep the nucleus shape but lose redundant colour channels, both simplifying the classical pipeline (Task 2) and the U-Net input (Task 3). The 6 training images are shown in RGB and grayscale in Fig. 1, the density and the sizes of the nuclei vary widely, and train_006 is sparse, while train_058 and train_034 are dense and close to each other, thus making both segmentation approaches used later difficult.

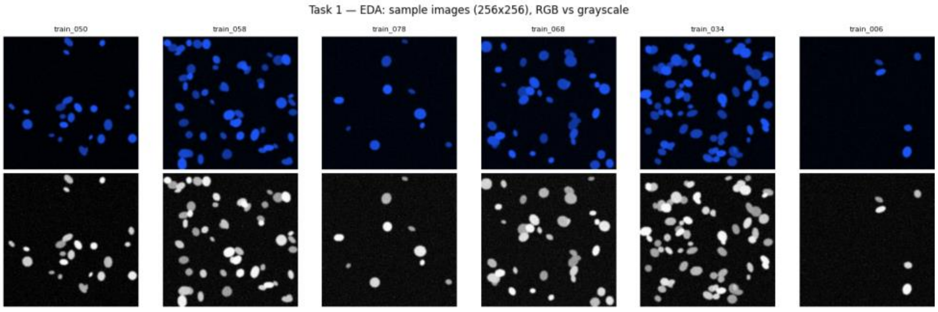

Figure 1. EDA sample grid: RGB (top) and grayscale (bottom) images showing variation in nucleus density and size.

The pooled pixel-intensity histogram for all the training images is shown in figure 2. It is heavily right skewed, with a lot of the pixels being close to 0 (black, or dark background) and a long thin tail of bright foreground pixels that are the bright objects in the image, such as the classic profile of a fluorescence image with a few bright objects on a dark background, which makes global Otsu thresholding (Task 2) a reasonable classical baseline for this data.

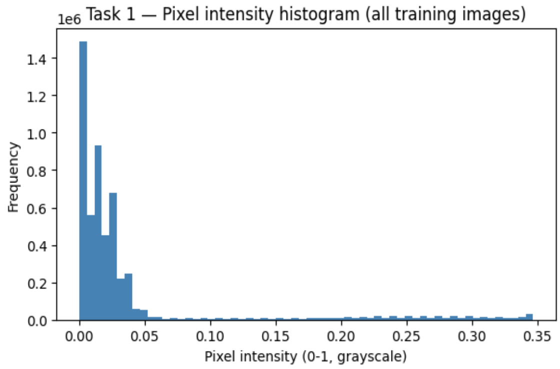

Figure 2. Pooled pixel-intensity histogram across all training images (0–1 grayscale).

Prompt engineering

When the prompt was a naive "What do you see in this image?," the responses were unconstrained and off-topic prose; for example, the live model said, "A starry night sky or perhaps an abstract pattern created for visual interest". There was no domain grounding nor any structure available for use in subsequent code. The following prompt was used instead:

You are an image-description assistant, assisting the preparation of data for a research pipeline, you are NOT a diagnostic tool and are never allowed to make a diagnosis, or to suggest that you are diagnosing disease – only describe what you are seeing. Respond strictly in JSON format with only these keys: modality, tissue_type, notable_features, and image_quality.

This makes the model descriptive rather than diagnostic, ensures an unchanging JSON schema (to prevent downstream code from parsing free text), and raises the bar on the diagnostic language, no matter what the image looks like. This was repeated twice on the model to examine the run-to-run stability (task1_vlm_outputs.json).

Note on this run: The Ollama application was on, minicpm-v was the default vision model and llama3.1 was the text model, and used_simulated_fallback was False. The two engineered-prompt runs of the same image were found to be different, with run 1 giving a report of "possibly a cellular sample or cell culture", and run 2 giving the report of "possibly lymphoid tissue". This shows a variation in the output of the live VLM from one run to the next.

2. Classical Features and LLM Interpretation (Task 2)

For the representative image (Figure 3), 9 objects were obtained from the image using Otsu thresholding followed by morphological opening and connected-component labelling and then the area, eccentricity, solidity, mean intensity, perimeter and extent were computed for each object using regionprops_table (Table 1; full table in task2_feature_table.csv).

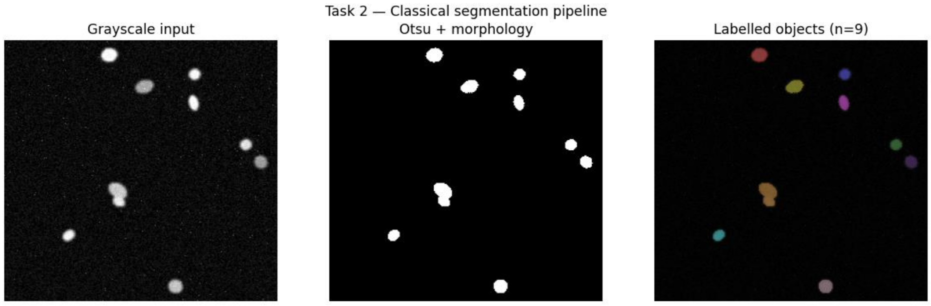

Figure 3. Classical pipeline: grayscale input → Otsu + morphology → labelled connected components (n=9).

Label	Area	Eccentr.	Solidity	Mean Int.	Extent
1	171.0	0.48	0.94	0.299	0.67
4	115.0	0.77	0.93	0.287	0.70
7	285.0	0.80	0.92	0.260	0.73
9	143.0	0.18	0.99	0.237	0.85

Table 1. Excerpt of the per-object regionprops feature table (4 of 9 objects shown).

This table was simplified to a text summary and given to the local LLM (llama3.1) — numbers only, no image. The model's output was the number of objects n_objects=9, density_class='uncertain', shape_regularity='regular' and quality_flag='good'. As seen in the direct VLM description, only qualitative observations were made and the descriptions of tissue types were inconsistent from one run to the next, whereas the numbers-first approach utilized the computed features directly, such as the verifiable mean intensity of 0.258. This is a way of helping with the numbers-first approach when definite quantities are needed (see Q1).

3. U-Net Segmentation (Task 3)

The supplied SmallUNet (base width 16) was trained using the combined BCE+Dice loss (selected a priori for stability and found to be justified only after the ablation in the Extension) at a resolution of 128×128 for 15 epochs. The training loss is seen to be smooth going from 1.31 to 0.66 as shown in figure 4, the validation Dice is going up sharply in the first 3 epochs 0.40 to 0.91 and it is seen to be flat range 0.91 to 0.94 as expected since it is a stricter non-linear mapping of the same overlap. The final validation metrics were Dice = 0.919, IoU = 0.851 (task3_unet_metrics.csv).

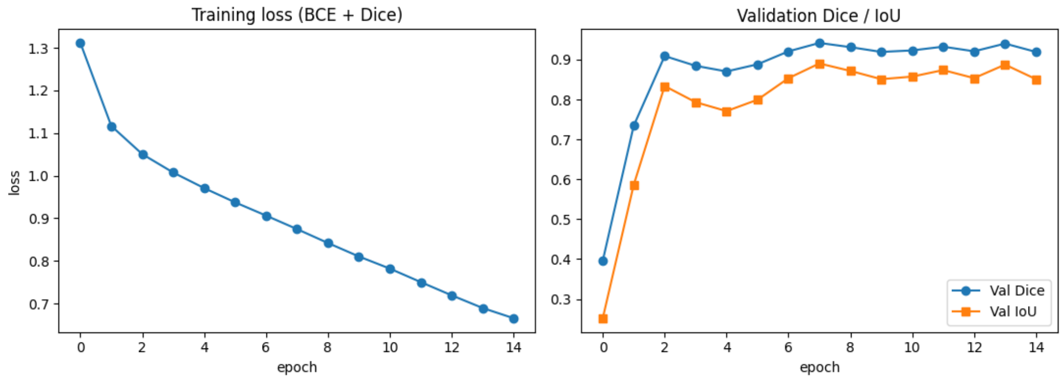

Figure 4. Training loss (BCE+Dice) and validation Dice/IoU per epoch.

Validation triplets (input, ground truth, prediction) are depicted in Figure 5. In general, the predictions demonstrate a good visual consistency with the ground-truth masks with some discrepancies in the resulting segmentation boundaries, and in smaller structures.

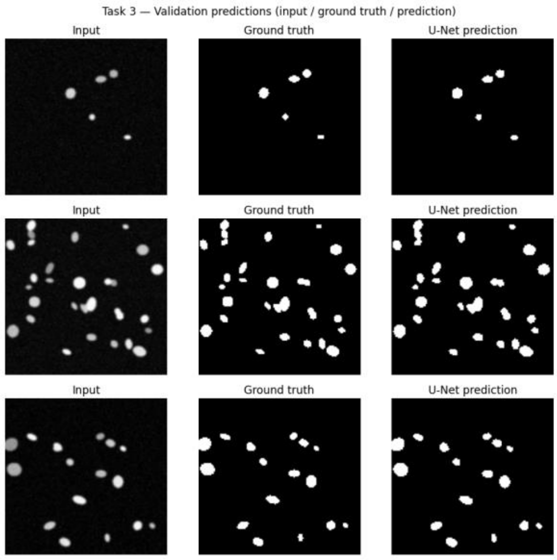

Figure 5. Validation triplets: input / ground truth / U-Net prediction.

4. Hybrid Pipeline (Task 4)

The trained U-Net was used for all 12 unseen test images, where the mean Dice and mean IoU were 0.9245 and 0.8597 respectively. The predicted masks were transformed into summaries at the image level, which were then given as input to llama3.1 to generate structured JSON records and narratives. Task4_test_records.csv was created by consolidating all 12 records. In one case, test_000, the following information was recorded: n_objects = 8, mean_area = 52.9 px, density_class = 'normal', quality_flag = 'good'.

Example narrative (test_000): "Objects are relatively evenly spaced and moderately sized indicating a typical or normal density of objects, the mean intensity is low, indicating that objects may appear faint or lightly coloured.

This is the same as Task 1 where the record and narrative were created by the actual Ollama (lama3.1) model rather than the simulated fallback. Each record contains the explicit flag used_simulated_fallback: False, to mark it for auditability. The same prompts and pipeline and downstream code is used with the same simulated earlier fall-back being replaced with the live model.

5. Extension: Loss Ablation and Robustness

Loss-function ablation

Loss ablation: BCE alone produced Dice = 0.894, Dice loss alone produced Dice = 0.156 (training essentially collapsed within 5 epochs), and the combination of BCE and Dice loss provided the greatest outcome, Dice = 0.909 (Figure 6). When forecasts began almost empty, dice loss by itself had trouble, but BCE offered a more robust per-pixel learning signal. As a result, combining BCE and Dice produced the best overlap performance and was chosen as Task 3's default loss.

Robustness to corruption

Test_000 has been subjected to high Gaussian blur (figure 7). In the raw image, it is very easy to see that there is blurring which makes the nuclei less distinct. Line 432, two adjacent nuclei that were segmented in clean image are now connected in the segmentation mask. This results in a decrease in the object count (8 to 7) and an increase in mean area (52.9 px to 117.3 px) in the downstream measurements. The LLM manages to generate a believable story, while not highlighting the segmentation mistake, underscoring the significance of structured JSON output to guarantee traceability of the results.

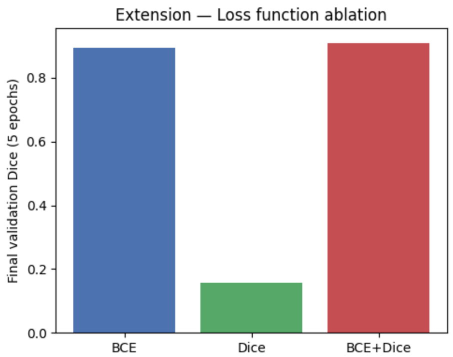

Figure 6. Validation-Dice loss ablation (BCE / Dice / BCE+Dice, 5 epochs).

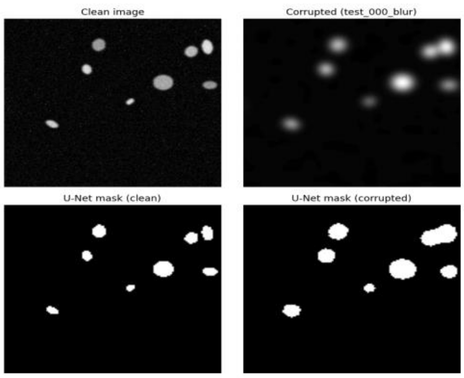

Figure 7. Clean vs. blur-corrupted image, and the corresponding U-Net masks.

Discussion Questions

Q1. Which description is more useful, and which is more trustworthy, the direct VLM description (Task 1) or the numbers-first description (Task 2) and why?

The numbers-first approach gives more trustworthy results for measurable quantities, since the results are a direct outcome of deterministic regionprops calculations, which are reproducible and auditable. The mean intensity for a given image is, for instance, always 0.258. However, the direct VLM is more helpful for more general impressions like overall image quality or unusual structures, but is less reproducible — two identical minicpm-v runs resulted in different tissue-type guesses: "cellular sample or cell culture" vs. "lymphoid tissue". Thus, the two approaches are complementary: When qualitative description is sufficient, use VLM and when some specific, reproducible measure is needed, use numbers-first.

Q2. Did the U-Net improve on classical Otsu segmentation for your modality? Give one example image where each approach did better.

For the clean separated training image (fig.3) it was sufficient to do classical segmentation with Otsu + morphology to get all 9 nuclei back, which is expected. The gap is opened up on the degraded input: For the blurred image in Figure 7, the U-Net mask yields compact, plausible nucleus-shaped regions (it has learned a shape before), while simply applying a global Otsu threshold would be much more apt to miss out low-contrast objects or connect noise into false blobs, since it lacks an idea of the shape to be expected. The U-Net's strength is therefore focused on the problem of imaging degradation, and not so much the high contrast and easy case.

Q3. Report your U-Net's Dice and IoU. What do these numbers mean, and where does the model tend to make its mistakes (which images or regions)?

The U-Net achieved a validation Dice of 0.919 and IoU of 0.851, and a test Dice of 0.9245 and IoU of 0.8597. Dice is the intersection of the segmentation mask of the foreground with the segmentation mask of the ground-truth, while IoU is the intersection of the foreground and ground-truth masks divided by their union and it is slightly more sensitive to segmentation inaccuracies. A strong level of agreement at the pixel level is shown by these scores. Main errors are the lack of detection of small or faint nuclei and close touching of these nuclei which sometimes lead to the nuclear area being merged (Figure 5).

Q4. Where in the pipeline can the LLM hallucinate, and what design choices reduce that risk? Why does keeping the structured JSON as the "source of truth" help?

The primary hazard is when the LLM is tasked with describing a picture or writing a story from images. May create specific measures, make unsupported diagnostic statements, or introduce information which is not backed by the results. This risk can be mitigated by using descriptive, non-diagnostic prompts, enforcing a fixed JSON schema, allowing values that go beyond a "yes" or "no" (such as "uncertain"), and pre-computing numerical summaries of the raw images in Tasks 2 and 4, and feeding them into the LLM. Having the structured JSON as the source of truth also allows the results to be auditable: The narrative should only describe the validated values in the structured JSON, and any disagreement can be identified and flagged.

Q5. Considering accuracy, auditability, and the limits of your dataset, would you trust any part of this system in a real clinical setting? What single change would most improve trustworthiness?

No. The U-Net was trained on a small dataset that is close to a synthetic dataset, without clinical validation or with no external test set. The live vision model also provided different results on equal inputs, indicating that outputs are not completely repeatable. There is no explicit measure of uncertainty from the pipeline at present. The most helpful improvement would be to incorporate a calibrated confidence score per image into the U-Net predictions (e.g. Monte-Carlo dropout) and use that to update the quality_flag. Low-confidence cases may then be automatically marked for human review as opposed to being presented with a high level of confidence as high-quality predictions would be.

References

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